Week 3 - Day 2

Library Catalog Database Design

1. Entities and Attributes

Authors

Attribute	Type
Id (PK)	UUID
Name	varchar
AuthorBiography	text

Translators

Attribute	Type
Id (PK)	UUID
Name	varchar
Language	varchar

Books

Attribute	Type
Id (PK)	UUID
Title	varchar
Price	decimal
Quantity	int
AuthorId (FK)	UUID
TranslatorId (FK)	UUID

Customers

Attribute	Type
Id (PK)	UUID
Name	varchar
Email	varchar
Password	varchar
Role	varchar

Orders

Attribute	Type
Id (PK)	UUID
Total	decimal
CustomerId (FK)	UUID

OrderItems

Attribute	Type
Id (PK)	UUID
Quantity	int
Subtotal	decimal
OrderId (FK)	UUID
BookId (FK)	UUID

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2. Normalization

1NF

- Each column should contain only one value.
- We should not store multiple values in the same column.
- Each row represents one record only.

Applied in this database:

Example:

Before applying 1NF:

Customers

Id	Name	Email	PhoneNumbers
1	Leen	leen@gmail.com	059xxxx, 056xxxx

The PhoneNumbers column contains multiple values in one field, which violates 1NF.

After applying 1NF:

Customers

Id	Name	Email
1	Leen	leen@gmail.com

CustomersPhones

PhoneId	CustomerId	PhonesNumber
1	1	059xxxx
2	1	056xxxx

Now each column contains a single value, and each record represents one piece of information.

2NF

- Every table should have a primary key.
- All other attributes in the table should depend on the whole primary key.
- A column should not depend on only part of the key.

Example before applying 2NF:

OrderItems table:

OrderId	BookId	BookTitle	Quantity
1	5	Book1	2
1	6	Book2	1

Primary key:

(OrderId + BookId)

Problem:

The BookTitle depends only on BookId, not on the whole primary key (OrderId + BookId).

After applying 2NF:

Books table:

BookId	BookTitle
5	Book1
6	Book2

OrderItems table:

OrderId	BookId	Quantity
1	5	2
1	6	1

Now every attribute depends on the complete primary key.

3NF

- Non-key columns should depend only on the primary key.
- A column should not depend on another non-key column.
- Avoid storing the same information in multiple table

Orders table:

OrderId	CustomerId	CustomerName	CustomerEmail
1	5	Leen	leen@gmail.com

Problem:

CustomerName and CustomerEmail depend on CustomerId, not on OrderId.

This causes duplicated data because the customer information will be repeated in every order.

After applying 3NF:

Customers table:

CustomerId	CustomerName	CustomerEmail
5	Leen	leen@gmail.com

Orders table:

OrderId	CustomerId
1	5

Now customer information is stored only once.

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3. Primary Keys and Foreign Keys

Primary Keys

- Authors.Id
- Translators.Id
- Books.Id
- Customers.Id
- Orders.Id

- OrderItems.Id

Foreign Keys

- Books.AuthorId → Authors.Id
- Books.TranslatorId → Translators.Id
- Orders.CustomerId → Customers.Id
- OrderItems.OrderId → Orders.Id
- OrderItems.BookId → Books.Id

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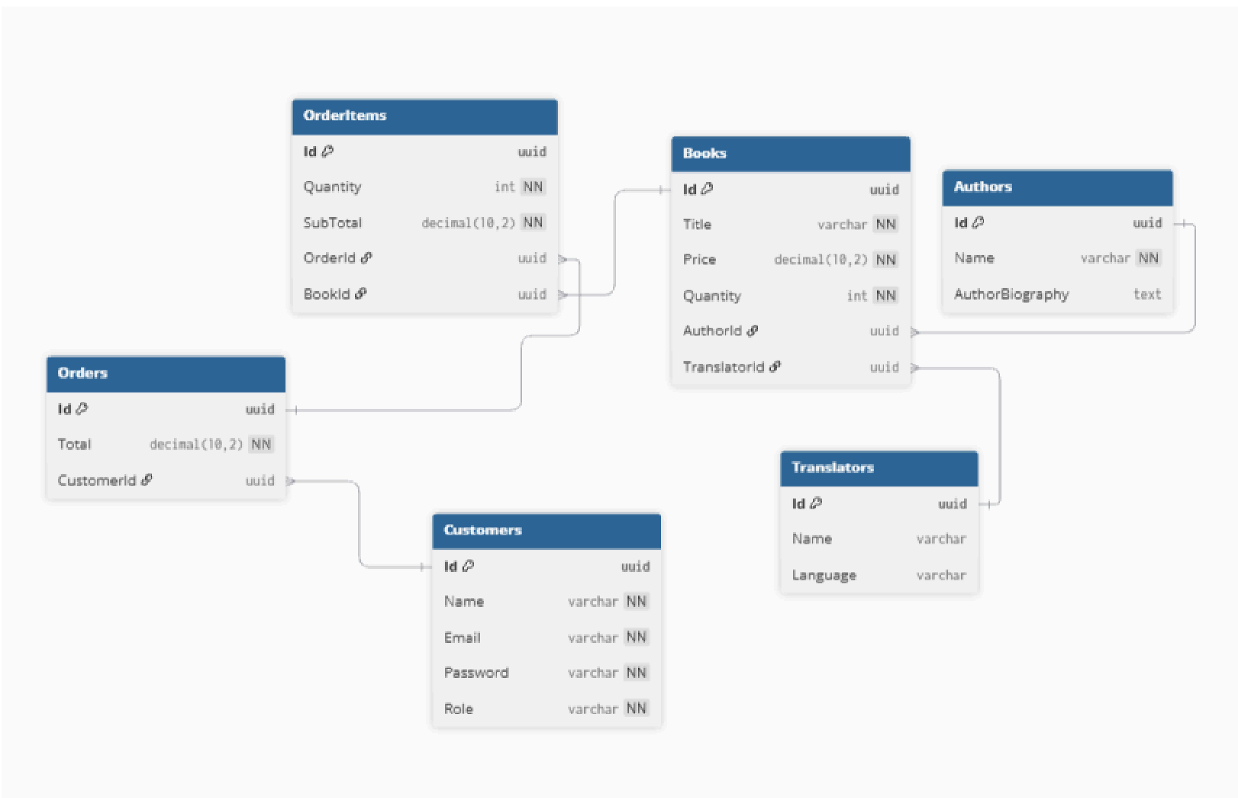
4. ERD Diagram

ERD was created using **dbdiagram.io**.



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5. Column Types

Type	Usage
UUID	Primary keys
varchar	Names, emails, roles, title , language ,password
text	Biography
decimal(10,2)	Monetary values (Price, Total, SubTotal)
int	Quantity

The following data types were selected based on the type of information stored in each attribute.

Column	Data Type	Reason
Id	UUID	Used as a unique identifier for each record
Name	varchar	Stores short text values such as names
Email	varchar	Stores email addresses
Password	varchar	Stores password values
Role	varchar	Stores user roles
Language	varchar	Stores translation language
AuthorBiography	text	Stores longer text information about authors
Price	decimal(10,2)	Used for monetary values to avoid floating-point rounding errors
Total	decimal(10,2)	Provides accurate calculations for order totals
Subtotal	decimal(10,2)	Provides accurate calculations for item prices
Quantity	int	Stores whole number values

Tools Used

- dbdiagram.io

ERD Diagram as PDF.

ERD_Digram.pdf